Unit VIII: Atoms and Nuclei

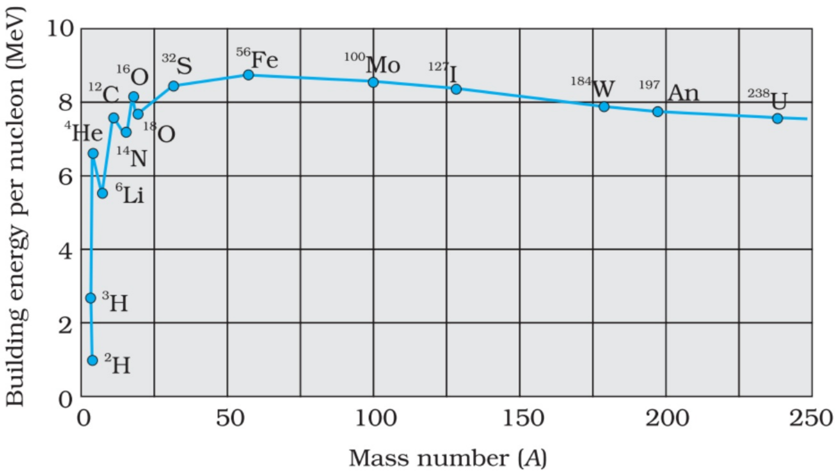
CHAPTER–13: NUCLEI

GIST OF CHAPTER:

Composition and size of nucleus, nuclear force , Mass-energy relation, mass defect; binding energy per nucleon and its variation with mass number; nuclear fission, nuclear fusion.

DEFINITIONS AND CONCEPTS:

1) **Composition of Nucleus:** Nucleus consists of two particles Protons and Neutrons. Protons are positively charged and Neutrons and neutral particles.
2) **Size of Nucleus:** Size of nucleus depends on number of nucleons (Protons and Neutrons). It increases with increase in number of nucleons.
   Size of nucleus $R = R_0 A^{1/3}$
3) **Nuclear force:** The force existing between nucleons (Proton – Proton, Neutron - Neutron, Proton – Neutron) inside the nucleus is called nuclear force. It is strongest force of in nature.
4) **Mass Defect:** Experimentally it has been found that the actual (Practical) mass of a nucleus is less than the expected theoretical mass of nucleus. This difference in Theoretical mass and practical mass is called as mass defect.
   Mass defect ΔM = Theoretical mass – Practical mass
   $\Delta M = [Zm_p + (A - Z) m_n] - M$
5) **Binding Energy:** If a certain number of neutrons and protons are brought together to form a nucleus of a certain charge and mass, an energy E_b will be released in the process. The energy E_b is called the binding energy of the nucleus.
   Binding Energy $E_b = \Delta Mc^2$
   Binding Energy per nucleon $E_{bn} = E_b/A$
   Binding energy per nucleon v/s Mass number



6) **Nuclear fission:** When neutron is bombarded on nucleus of heavy elements like uranium, it breaks into two nuclei. This process of breaking nucleus into fragments is called Nuclear fission.
7) **Nuclear fusion:** When two light nuclei combine together to form a larger nucleus, energy is released in this process. This is called as nuclear fusion.
8) **Q-Value:** Energy liberated during fission reaction is called as Q - value.

FORMULAE

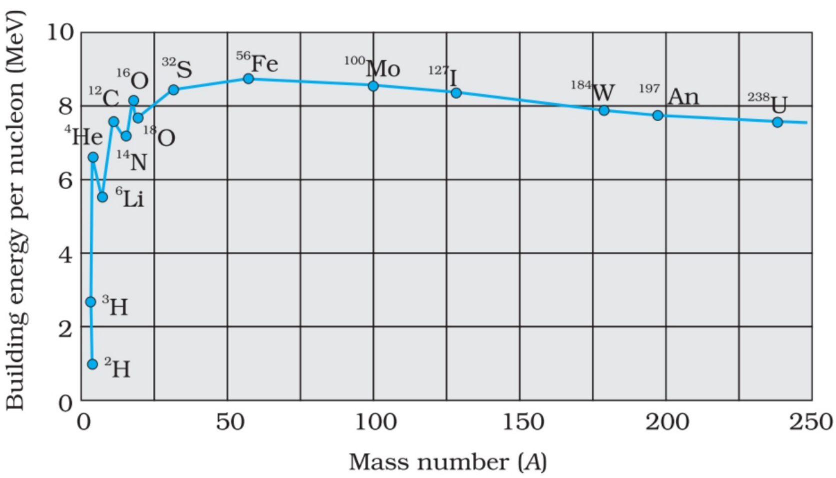
- Size of nucleus — $R = R_0 A^{1/3}$
- Mass energy relation — $E = mc^2$
- Mass defect — ΔM = Theoretical mass – Practical mass
  $\Delta M = [Zm_p + (A - Z)m_n] - M$
- Binding Energy — $E_b = \Delta Mc^2$
- Binding Energy per nucleon — $E_{bn} = E_b/A$

UNITS

- Size of nucleus — m (metre) or fm (fermi or fempto metre)
- Mass defect — kg (kilogram)or u (unified mass) (1 u = 1.660563×10^{-27} kg)
- Binding Energy — J (joule) or MeV (mega electron volt)
- Binding Energy per nucleon — J (joule) or MeV (mega electron volt)

GRAPHS

- Binding energy per nucleon v/s Mass number



- Potential energy of pair of nucleons as a function of separation between them